

Digital Logic Design – Assignment

Part A: Conceptual Questions

1. Explain the difference between combinational circuits and sequential circuits with examples.
2. Convert the following numbers: (a) $(101101.011)_2$ to decimal (b) $(245)_{10}$ to binary (c) $(7A)_{16}$ to binary.
3. State and explain De Morgan's Theorems with proof.
4. Compare SR Flip-Flop, JK Flip-Flop, and D Flip-Flop.
5. What is a race around condition? How can it be eliminated?

Part B: Problem Solving

6. Simplify the Boolean function using K-map: $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$.
7. Design a half adder and full adder: Write truth tables, derive Boolean expressions, and draw logic diagrams.
8. Implement the Boolean function using NAND gates only: $F = A'B + AB'$.
9. Design a 3-to-8 decoder and explain its working.
10. A sequential circuit has one input X and one output Z. The output becomes 1 whenever the input sequence '101' is detected. Design the state diagram and state table.

Part C: Design/Application

11. Design a mod-6 synchronous counter using JK flip-flops.
12. Design a 4-bit shift register and explain its operation.
13. Implement a multiplexer-based realization of the function: $F(A, B, C) = \sum m(1, 2, 6, 7)$.
14. Write short notes on: (a) PLA (b) PAL (c) FPGA.
15. Design a simple digital lock system using basic logic gates.

Bonus (Optional)

16. Design a traffic light controller using sequential logic.
17. Explain the difference between Mealy and Moore machines with examples.